COURSE NLP301 / MODULE NL

NLP Models and Tasks

Learning goal: explain core concepts, compare related methods, and apply them to realistic machine learning or educational technology scenarios.

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01. How is text classification evaluated?

Text classification assigns labels such as topic or sentiment. Use a held-out set and inspect precision, recall, F1, and confusion matrices, especially for imbalanced classes and classes with overlapping language. In practice, it helps to think of this as a sequence of steps rather than a single operation. Each step affects the information available to the next stage, so implementation choices should be validated on representative data. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

02. What is sentiment analysis?

Sentiment analysis estimates positive, negative, or neutral attitude expressed in text. Sarcasm, mixed opinions, and domain language make labels uncertain; a model trained on reviews may fail on student feedback. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

03. What is named entity recognition?

NER identifies spans such as people, organizations, dates, or course names. Evaluation should account for both correct category and exact span boundaries. Labels must be defined consistently in annotation guidelines. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

04. What is part-of-speech tagging?

POS tagging assigns grammatical categories to words in context, such as noun or verb. Ambiguous words need their neighboring words to resolve their role; tagging can support downstream parsing or extraction. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

05. How do extractive and generative question answering differ?

Extractive QA selects a text span from a given passage. Generative QA composes a response that may combine passages. Generative systems need evidence checks because fluent answers can introduce unsupported claims. A useful way to understand how do extractive and generative question answering differ is to compare the methods not only by definition, but also by the kind of input they expect, the output they produce, and the situations in which each is useful. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

06. What is text summarization?

Summarization condenses a source while preserving its main points. Extractive methods select existing sentences; abstractive methods rewrite them. Evaluate whether important facts survive and whether added claims are supported. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

07. When might an LSTM be used for NLP?

An LSTM can classify or tag token sequences using learned recurrent states and embeddings. For modest datasets and constrained inference it may be useful, though pretrained transformers often outperform it on broad language tasks. The choice is context dependent. The right approach depends on the objective, the data distribution, the cost of errors, and operational constraints such as latency, memory, interpretability, and the amount of labeled data available. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.